# A COUNTEREXAMPLE, WITH A HYPERBOLIC SUMMAND, TO A PRINTED SUBADDITIVITY QUESTION FOR TOPOLOGICAL VOLUME

ABSTRACT. For a closed orientable 3–manifold $M$, Kegel, Ray, Spreer, Em Thompson and Tillmann define the *topological volume* $\mathrm{vol_t}(M)$ to be the infimum of $\mathrm{vol_h}(M \setminus L)$ over the hyperbolic links $L \subset M$, the empty link included, and ask whether $\mathrm{vol_t}(M\#N) \leq \mathrm{vol_t}(M) + \mathrm{vol_t}(N)$ for all closed orientable $M, N$. We exhibit one pair for which this fails, so the question is false exactly as printed. Let $W$ be the Weeks manifold. Then $W$ is hyperbolic, so $\mathrm{vol_t}(W) = \mathrm{vol_h}(W) \leq 0.943$, whereas $W\#W$ is reducible and therefore not hyperbolic, so every competitor in its infimum is an *orientable cusped* finite-volume hyperbolic 3–manifold and Cao–Meyerhoff gives $\mathrm{vol_t}(W\#W) \geq 2v_0 = 2.029883212819307\ldots > 1.886 \geq \mathrm{vol_t}(W) + \mathrm{vol_t}(W)$. Both summands here are hyperbolic; the reading of the question in which the summands are not hyperbolic, which is the substantive connected-sum problem, is untouched.

## 1. The question

Let $M$ be a closed orientable 3–manifold. A *link* in $M$ is a possibly empty, possibly disconnected closed 1–submanifold $L \subset M$, and $L$ is *hyperbolic* if $M \setminus L$ admits a complete hyperbolic structure of finite volume. Kegel, Ray, Spreer, Em Thompson and Tillmann [4] define

$$\mathrm{vol_t}(M) = \inf\{\, \mathrm{vol_h}(M \setminus L) \mid L \subset M \text{ is a hyperbolic link} \,\}, \tag{1}$$

following Bessières, Besson, Boileau, Maillot and Porti [1]. Because the empty link is admitted, they record immediately after (1) that

$$\mathrm{vol_t}(M) = \mathrm{vol_h}(M) \qquad \text{whenever } M \text{ is hyperbolic}, \tag{2}$$

in their own words: *"If $M$ admits a complete hyperbolic structure, then* $\mathrm{vol_t}(M) = \mathrm{vol_h}(M)$*, since volume decreases under Dehn filling."* Write $v_0$ for the volume of the regular ideal hyperbolic tetrahedron, so that

$$v_0 = 1.0149416064096536250212 03\ldots,$$
$$2v_0 = 2.0298832128193072500424 05\ldots$$

In the subsection *Connected sums* of [4] the authors ask the following, quoted verbatim.

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**Question** ([4], first Question of the subsection *Connected sums*)**.** “Let $M$ and $N$ be closed, orientable 3-manifolds. Is $\mathrm{vol}_{\mathrm{t}}(M\#N) \leq \mathrm{vol}_{\mathrm{t}}(M) + \mathrm{vol}_{\mathrm{t}}(N)$?”

As printed, the Question carries exactly two hypotheses, *closed* and *orientable*: no primeness, no irreducibility and no hyperbolicity clause. We found no standing convention elsewhere in [4] that would add one.

## 2. Two lemmas

**Lemma 1** (a floor at non-hyperbolicity)**.** *Let $X$ be a closed orientable 3–manifold that is not hyperbolic. Then* $\mathrm{vol}_{\mathrm{t}}(X) \geq 2v_0$*.*

*Proof.* Let $L \subset X$ be a hyperbolic link. Then $L \neq \emptyset$: otherwise $X$ itself would carry a complete finite-volume hyperbolic structure, contrary to hypothesis. Hence $X \setminus L$ is a non-compact — so cusped — complete finite-volume hyperbolic 3–manifold, and it is orientable, orientability being inherited from $X$. By Cao–Meyerhoff [2], *every orientable cusped hyperbolic 3–manifold has volume at least* $2v_0$, whence $\mathrm{vol}_{\mathrm{h}}(X \setminus L) \geq 2v_0$. Take the infimum over $L$. □

*Remark.* The hypothesis *orientable* in [2] is load-bearing: the Gieseking manifold is a non-orientable cusped hyperbolic 3–manifold of volume $v_0 = 1.014941606409653\ldots < 1.886$, so a floor of $v_0$ would not suffice below. Lemma 1 does not presuppose that the infimum in (1) is attained, nor that it is finite: every competitor is bounded below by $2v_0$, so the infimum is, whatever it is.

**Lemma 2** (reducible $\Rightarrow$ not hyperbolic)**.** *If neither $M$ nor $N$ is $S^3$, then $M\#N$ is reducible and is not hyperbolic.*

*Proof.* The summing sphere bounds a ball on neither side, so it is essential and $M\#N$ is reducible. A closed hyperbolic 3–manifold has universal cover $\mathbb{H}^3$, hence is aspherical, hence has $\pi_2 = 0$, hence contains no essential embedded 2–sphere by the Sphere Theorem: it is irreducible. □

## 3. The counterexample

Let $W$ be the Weeks manifold. It is closed, orientable and hyperbolic, so $\mathrm{vol}_{\mathrm{t}}(W) = \mathrm{vol}_{\mathrm{h}}(W)$ by (2), and

$$\mathrm{vol}_{\mathrm{h}}(W) \leq 0.943 \tag{3}$$

by Gabai–Meyerhoff–Milley, whose abstract reads, verbatim: *“We also show how this result can be used to construct a complete list of all one-cusped hyperbolic three-manifolds with volume* $\leq 2.848$ *and all closed hyperbolic three-manifolds with volume* $\leq 0.943$*. In particular, the Weeks manifold is the unique smallest volume closed orientable hyperbolic* 3*-manifold.”* We use only the *bound* (3), never a decimal expansion of $\mathrm{vol}_{\mathrm{h}}(W)$.

**Theorem 3.** $\mathrm{vol_t}(W\#W) \geq 2v_0 > 1.886 \geq \mathrm{vol_t}(W) + \mathrm{vol_t}(W)$. *In particular the Question of §1 is false at the pair* $(W, W)$.

*Proof.* By (2) and (3), $\mathrm{vol_t}(W) + \mathrm{vol_t}(W) \leq 2 \times 0.943 = 1.886$. Neither summand is $S^3$, so $W\#W$ is not hyperbolic by Lemma 2, and Lemma 1 applied to $X = W\#W$ gives $\mathrm{vol_t}(W\#W) \geq 2v_0$. Finally $2v_0 - 1.886 \geq 0.143883212819307 > 0$. $\square$

## 4. What is not settled

Both summands of the counterexample are hyperbolic, and that is what the argument uses: the substantive connected-sum problem, in which the summands are not hyperbolic, is untouched. There Lemma 1 already gives $\mathrm{vol_t}(M), \mathrm{vol_t}(N) \geq 2v_0$, so a counterexample would need a lower bound on $\mathrm{vol_t}(M\#N)$ exceeding a sum of at least $4v_0$, and nothing above supplies one. The authors of [4] state (2) and quote Cao–Meyerhoff's $2v_0$ minimum themselves, so a hypothesis of primeness, or of non-hyperbolic summands, may well have been intended and omitted; we settle the question as printed and make no claim about what was meant. Nothing above supports or refutes $\mathrm{vol_t}(M\#N) \leq \mathrm{vol_t}(M) + \mathrm{vol_t}(N) + 2v_0$ or any other repaired form. We exhibit the one pair $(W, W)$ and count no others. The two mathematical inputs beyond (3) are the source's own: (2) is its sentence, and it quotes Cao–Meyerhoff itself.

**Adjacent work.** The subsection *Connected sums* of [4] records

$$\mathrm{vol_t}(L(2,1)\#L(3,1)) < \mathrm{vol_t}(L(2,1)) + \mathrm{vol_t}(L(3,1)),$$

an instance in which the inequality of the Question of §1 holds strictly, and the summary of related work in [4] says on the strength of it that $\mathrm{vol_t}$ "is not additive under connected sum"; that sentence is about that strict inequality and not about any failure of the inequality it asks about, which [4] leaves open. Example 2.1 of [4] computes $\mathrm{vol_t}(S^3) = 2v_0$ from the same minimum-volume input, which is the argument of Lemma 1 run for the single manifold $S^3$; we did not find the general statement of Lemma 1 there. The Oberwolfach report [5], by a working group including a co-author of [4], closes by asking to better understand the behaviour of hyperbolic volume under connected sums, and settles nothing about the inequality above.

## 5. Verification

The result above is a hand proof: two lemmas from published theorems, then arithmetic on three decimal places. Nothing in §§1–3 needs a computer.

A program (`verify.py`, with its transcript `verify.output.txt`) accompanies this note because the note prints numbers. It is standard-library Python, and in exact arithmetic it recomputes $v_0$ and $2v_0$ from $2v_0 = 3\,\mathrm{Cl}_2(2\pi/3)$ by the Clausen–Bernoulli expansion with an explicit truncation bound, cross-checked against $v_0 = \mathrm{Cl}_2(\pi/3)$ via the Clausen duplication identity, and re-decides $2v_0 > 1.886$ and $v_0 < 1.886$ as comparisons of exact rationals. Those digit expansions are the only values in this note that come from computation rather than from print; the bound 0.943 is transcribed, and no decimal expansion of $\mathrm{vol}_{\mathrm{h}}(W)$ is used anywhere. The program does not, and cannot, check the topological content quoted from the literature — (2), Cao–Meyerhoff, Gabai–Meyerhoff–Milley, the Sphere Theorem and Kneser–Milnor are imported, and its own transcript says so.

The transcript's 38 checks also cover material this note does not state, and they are not evidence for anything it does state: rows of Table 1 of [4]; a second route through a census reach of 3.07 quoted from a theorem this note does not use, together with ten further pairs $W\#N$ built on that route, whose premise — irreducibility of those rows — the run leaves unverified; by-product ratios; and one floating-point SnapPy measurement that the run itself flags as not interval-certified. Theorem 3 uses none of them, and this note asserts none of them.

## References


[1] L. Bessières, G. Besson, M. Boileau, S. Maillot, and J. Porti, *Geometrisation of* 3-*manifolds*, EMS Tracts in Mathematics **13**, European Mathematical Society, 2010, §1.3.2. doi:10.4171/082. Not open access; consulted only through works that restate its definition of the invariant, including [4].

[2] C. Cao and G. R. Meyerhoff, *The orientable cusped hyperbolic* 3-*manifolds of minimum volume*, Invent. Math. **146** (2001), no. 3, 451–478. doi:10.1007/s002220100167; MR1869847. The hypothesis *orientable* in the title is the one used in Lemma 1.

[3] D. Gabai, R. Meyerhoff, and P. Milley, *Minimum volume cusped hyperbolic three-manifolds*, arXiv:0705.4325. The bound (3) is the sentence of the abstract quoted in §3, read at that address; that quotation, and not a journal volume or page, is the citation this note relies on. We give no journal coordinates for this item because we did not check any.

[4] M. Kegel, A. Ray, J. Spreer, E. Thompson, and S. Tillmann, *On a volume invariant of* 3-*manifolds*, Expo. Math. **44** (2026), no. 4, Paper No. 125782, 31 pp. doi:10.1016/j.exmath.2026.125782; preprint arXiv:2402.04839v2. All quotations in this note are from the `v2` e-print, the version revised following a referee report.

[5] K. Ben Aribi, B. Burton, S. Ibarra, J. Morgan, J. Purcell, J. S. Quintanilha, F. Santoro, S. Schleimer, and E. Thompson, *Computing the topological volume of some three-manifolds*, Oberwolfach Rep. **21** (2024), no. 3, 2530–2534. doi:10.4171/OWR/2024/43. *Caution:* the coordinates [4] gives for this report do not resolve; the item is identified here by title and author list.